FROB C5 characteristic-five countercontrol

Validation audit, 22 October 2026

Question. What happens to the coprime-factor kernel argument used for the archived $q = 11, 13, 19$ cells if the characteristic equals the Frobenius order five? This is a mathematical negative control, not a new run.

Lean result. Over $\mathbf{F}_5$ the source proves

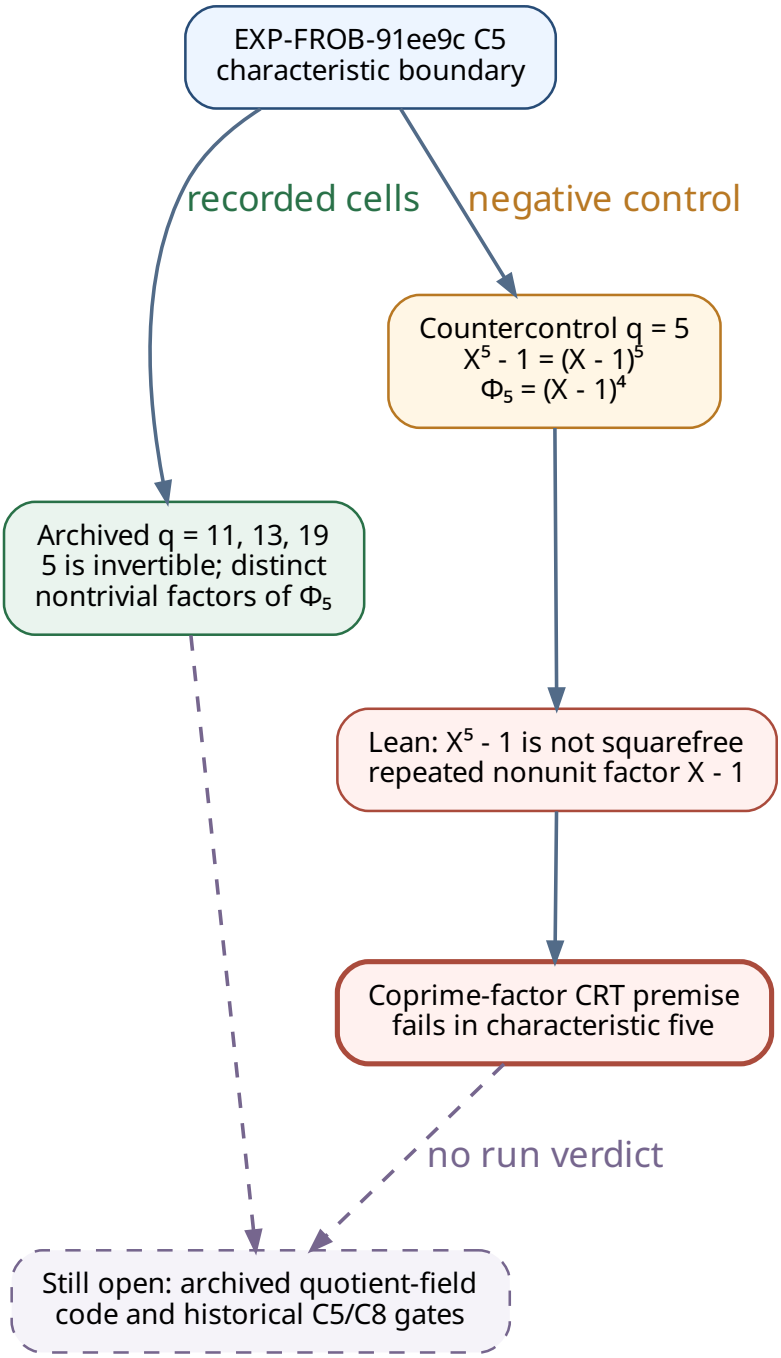
$$X^5 - 1 = (X - 1)^5, \qquad X^4 + X^3 + X^2 + X + 1 = (X - 1)^4.$$

Consequently $X^5 - 1$ is not squarefree: the nonunit $X - 1$ has a square dividing it. The exact target theorem is `CharacteristicFive.x_pow_five_sub_one_not_squarefree`.

Proof method. Mathlib's characteristic-five binomial identity establishes the fifth-power equation. Ring normalization and checked $\mathbf{F}_5$ coefficient identities establish the degree-four equation. The squarefreeness definition then contradicts the degree-one, nonunit divisor $X - 1$. The Lean source commit begins `203cff284eff`; the attached verifier receipt binds that source and pinned Lean 4.35.0-rc1. The 8,971-job workspace and 3,092-job target builds passed; 588 theorem declarations used standard axioms only, and the target axiom audit passed. Independent semantic review remains pending.

Interpretation. The coprime-factor CRT premise fails at characteristic five. This does not contradict the q=11/13/19 mathematical identities in companion PR #2475. The archived trial contains no q=5 core cell. Its quotient-field implementation, historical C5/C8 verdicts, source custody and performance claims are not certified by this theorem.

Graphs. The diagram on page two is a boundary map. No numerical performance graph or dashboard row changes because no measurement was made.


EXP-FROB-91ee9c C5
characteristic boundary
recorded cells
negative control
Countercontrol q = 5
$X^5 - 1 = (X - 1)^5$
$\Phi_5 = (X - 1)^4$
Archived q = 11, 13, 19
5 is invertible; distinct
nontrivial factors of Φ_5
Lean: $X^5 - 1$ is not squarefree
repeated nonunit factor X - 1
Coprime-factor CRT premise
fails in characteristic five
no run verdict
Still open: archived quotient-field
code and historical C5/C8 gates